

## MONT-RIGI, Belgium

WMO: 064940

Lat: 50.5108N

Lon: 6.0733E

Elev: 673

StdP: 93.50

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|------------|-------------|------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |            | 1%          |            |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-9.9 | (c)<br>-8.0 | (d)<br>-14.4                  | (e)<br>1.2 | (f)<br>-7.0 | (g)<br>-10.8 | (h)<br>1.6 | (i)<br>-7.1 | (j)<br>11.3           | (k)<br>4.5 | (l)<br>10.4 | (m)<br>4.0 | (n)<br>2.6            | (o)<br>80 | (p)<br>0.623 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.7                    | 26.5            | 18.2 | 24.1 | 17.1 | 22.2 | 16.0 | 19.0                | 24.3 | 17.9 | 22.6 | 16.9 | 20.8 | 3.3                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.1                            | 13.3 | 20.5 | 16.2 | 12.5 | 19.5 | 15.3 | 11.8 | 18.5 | 56.7          | 24.5 | 52.9 | 22.9 | 50.0 | 21.1 | 24.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 9.7               | 8.6  | 7.6 | -12.3                      | 30.3 | 2.9                | 1.8 | -14.4                                              | 31.6 | -16.0      | 32.7 | -17.7      | 33.7 | -19.8      | 35.0 |
|                   |      |     | -13.0                      | 21.7 | 2.9                | 1.2 | -15.1                                              | 22.6 | -16.8      | 23.3 | -18.5      | 23.9 | -20.6      | 24.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 7.4    | -0.7  | -0.3  | 3.2   | 6.9   | 10.3  | 13.3  | 15.4  | 15.1  | 11.7  | 8.4   | 4.1   | 0.8  |
|                                                                                  | DBStd       | 6.82   | 3.59  | 4.38  | 4.17  | 4.30  | 4.07  | 3.65  | 3.68  | 3.31  | 3.11  | 3.90  | 3.66  | 3.84 |
|                                                                                  | HDD10.0     | 1558   | 332   | 289   | 214   | 112   | 47    | 8     | 1     | 2     | 14    | 78    | 178   | 284  |
|                                                                                  | HDD18.3     | 4024   | 590   | 522   | 470   | 342   | 249   | 156   | 107   | 113   | 199   | 308   | 426   | 542  |
|                                                                                  | CDD10.0     | 613    | 0     | 1     | 3     | 20    | 57    | 108   | 168   | 159   | 66    | 28    | 2     | 0    |
|                                                                                  | CDD18.3     | 36     | 0     | 0     | 0     | 0     | 1     | 6     | 16    | 12    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 277    | 0     | 0     | 0     | 0     | 7     | 52    | 119   | 91    | 8     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 55     | 0     | 0     | 0     | 0     | 1     | 7     | 30    | 17    | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.8    | 4.3   | 4.3   | 4.1   | 3.4   | 3.6   | 3.1   | 3.2   | 3.1   | 3.2   | 3.9   | 4.4   | 4.5  |
| Precipitation                                                                    | PrecAvg     | 1026   | 97    | 87    | 76    | 63    | 74    | 86    | 91    | 94    | 80    | 79    | 90    | 110  |
|                                                                                  | PrecMax     | 1244   | 202   | 230   | 160   | 138   | 171   | 174   | 223   | 202   | 204   | 209   | 151   | 266  |
|                                                                                  | PrecMin     | 799    | 6     | 17    | 12    | 0     | 33    | 27    | 28    | 39    | 16    | 34    | 3     | 21   |
|                                                                                  | PrecStd     | 126    | 47    | 46    | 42    | 31    | 29    | 34    | 43    | 47    | 49    | 37    | 35    | 51   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 8.4   | 13.8  | 16.9  | 20.5  | 24.3  | 27.7  | 30.1  | 28.7  | 24.6  | 20.7  | 14.9  | 9.7  |
|                                                                                  |             | MCWB   | 8.1   | 6.0   | 8.7   | 11.1  | 16.0  | 19.0  | 18.9  | 18.9  | 17.3  | 14.1  | 10.8  | 8.7  |
|                                                                                  | 2%          | DB     | 6.8   | 8.3   | 14.3  | 18.6  | 21.3  | 24.6  | 26.5  | 26.1  | 21.5  | 18.0  | 12.0  | 8.4  |
|                                                                                  |             | MCWB   | 6.4   | 5.4   | 8.0   | 10.8  | 13.7  | 17.3  | 18.2  | 18.0  | 15.6  | 13.8  | 9.7   | 7.6  |
|                                                                                  | 5%          | DB     | 5.4   | 6.5   | 11.3  | 16.2  | 19.3  | 22.2  | 23.9  | 23.2  | 19.2  | 15.2  | 10.5  | 7.2  |
|                                                                                  |             | MCWB   | 4.9   | 5.1   | 6.6   | 10.1  | 13.0  | 15.9  | 17.1  | 17.1  | 14.8  | 12.5  | 9.3   | 6.4  |
|                                                                                  | 10%         | DB     | 4.1   | 5.1   | 8.7   | 13.7  | 17.4  | 19.9  | 21.9  | 20.9  | 16.8  | 13.4  | 9.1   | 5.9  |
|                                                                                  |             | MCWB   | 3.6   | 4.1   | 5.8   | 8.9   | 12.4  | 15.1  | 16.1  | 15.7  | 13.7  | 11.6  | 8.3   | 5.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.3   | 7.7   | 9.7   | 12.6  | 17.6  | 20.3  | 20.2  | 20.2  | 17.9  | 15.5  | 12.5  | 9.2  |
|                                                                                  |             | MCDB   | 8.3   | 10.8  | 14.2  | 18.1  | 22.0  | 26.3  | 26.5  | 26.5  | 23.0  | 18.5  | 13.4  | 9.4  |
|                                                                                  | 2%          | WB     | 6.5   | 6.5   | 8.5   | 11.5  | 15.6  | 18.0  | 18.9  | 18.8  | 16.5  | 14.2  | 10.4  | 7.9  |
|                                                                                  |             | MCDB   | 6.7   | 7.7   | 12.5  | 16.6  | 19.0  | 22.8  | 24.5  | 23.8  | 20.2  | 16.7  | 11.4  | 8.2  |
|                                                                                  | 5%          | WB     | 5.0   | 5.3   | 7.4   | 10.4  | 14.1  | 16.7  | 17.9  | 17.6  | 15.2  | 13.0  | 9.4   | 6.6  |
|                                                                                  |             | MCDB   | 5.3   | 6.5   | 10.4  | 14.8  | 17.8  | 20.7  | 23.1  | 21.6  | 18.6  | 15.0  | 10.3  | 7.1  |
|                                                                                  | 10%         | WB     | 3.8   | 4.0   | 6.3   | 9.3   | 12.8  | 15.6  | 16.8  | 16.6  | 14.1  | 12.0  | 8.3   | 5.3  |
|                                                                                  |             | MCDB   | 4.0   | 4.9   | 8.4   | 13.2  | 16.4  | 19.2  | 20.7  | 20.1  | 16.4  | 13.3  | 9.1   | 5.8  |
| Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 3.5   | 4.4   | 6.3   | 8.2   | 8.5   | 8.6   | 8.7   | 8.7   | 7.5   | 6.0   | 4.3   | 4.0  |
|                                                                                  | 5% DB       | MCDBR  | 4.6   | 6.7   | 10.5  | 11.1  | 11.0  | 11.7  | 11.7  | 11.9  | 10.3  | 8.8   | 6.6   | 5.6  |
|                                                                                  |             | MCWBR  | 4.5   | 5.0   | 5.9   | 5.7   | 5.8   | 6.3   | 5.8   | 6.2   | 5.9   | 5.5   | 5.0   | 4.6  |
|                                                                                  | 5% WB       | MCDBR  | 4.4   | 5.9   | 8.7   | 10.1  | 9.4   | 10.4  | 10.2  | 10.2  | 9.3   | 7.5   | 5.8   | 4.8  |
|                                                                                  |             | MCWBR  | 4.4   | 4.8   | 5.7   | 5.6   | 5.7   | 6.3   | 5.7   | 6.0   | 5.6   | 5.2   | 4.9   | 4.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.290  | 0.311 | 0.341 | 0.372 | 0.379 | 0.381 | 0.382 | 0.376 | 0.349 | 0.330 | 0.307 | 0.287 |      |
|                                                                                  | taud        | 2.465  | 2.418 | 2.369 | 2.311 | 2.321 | 2.327 | 2.317 | 2.360 | 2.421 | 2.479 | 2.478 | 2.446 |      |
|                                                                                  | Ebn at Noon | 760    | 829   | 861   | 869   | 877   | 876   | 868   | 856   | 843   | 795   | 728   | 702   |      |
|                                                                                  | Edh at Noon | 58     | 79    | 99    | 117   | 121   | 121   | 121   | 110   | 93    | 73    | 58    | 52    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.77   | 1.45  | 2.59  | 4.12  | 4.86  | 5.33  | 5.05  | 4.33  | 3.24  | 1.90  | 0.89  | 0.60  |      |
|                                                                                  | RadStd      | 0.12   | 0.28  | 0.38  | 0.48  | 0.47  | 0.50  | 0.60  | 0.44  | 0.38  | 0.18  | 0.12  | 0.09  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (74 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page